

Drum Track Separation

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Abstract—An algorithm is proposed to separate out the audio from a recording of a drum kit, corresponding to the audio produced from different drums in the kit. The three drum types taken into consideration in this system are the kick drum, snare drum, and high hat. The system highly utilizes and adapts the Non-Negative Matrix Factorization (NMF) algorithm. The success of this algorithm was assessed with regard to its ability to identify the correct drum being played at a given time. The annotated dataset IDMT-SMT-Drums was used to generate this performance assessment. The algorithm succeeded in identifying drum hits and separating the kit into individual drum tracks. However, the output separated audio tracks were significantly corrupted when two drums were played concurrently.

I. INTRODUCTION

A. Context

In the fields of music recording and live music performance, the audio reinforcement of drums is often a uniquely difficult task. This is in part due to the large number of audio sources in a standard western drum kit, each of which often warrant its own dedicated microphone and processing before being added back into an audio mix. Because processing is often done on a single drum at a time, being able to take a recording of an entire drum kit and break it down into its different drum source components could allow a producer or sound engineer the same flexibility in processing, while requiring fewer audio pickups. Drum separation might also prove useful if a producer only has access to a mixed drum kit recording, but wants to alter only the sound from a single drum. For these reasons, the problem of drum track separation was investigated.

B. Problem Statement

The task consists of taking a recording of a drum kit and separating out the audio tracks corresponding to the audio components resulting from each of the different drum types being played. For this specific investigation, the input drum audio was composed of sounds from three types of drums: the high-hat, snare drum, and kick drum. These three components are some of the most commonly employed on a standard drum kit and have significantly distinct frequency spectra, which was anticipated to enable more accurate identification given the approach proposed below.

II. SYSTEM DESIGN

A. NMF Overview

Drum track separation was performed by applying NMF to the Short Time Fourier Transform (STFT) of the input audio recording. The NMF algorithm takes in an input V matrix and

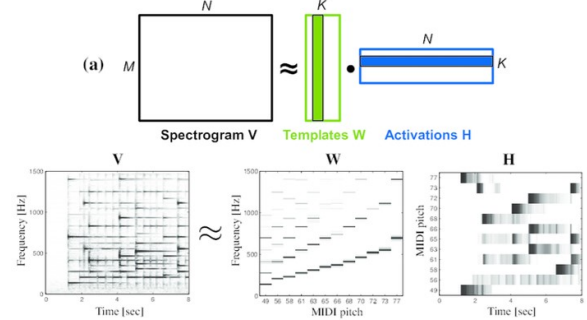


Fig. 1. NMF overview[1]

and attempts to break it down into two non-negative matrices H and W .

$$V \approx HW$$

Input initial guesses of H and W are also supplied to the NMF algorithm. H and W are then updated a specified number of times, minimizing the Frobenius norm equation as shown below.

$$\|V - WH\|$$

In this context, the W matrix supplied to the algorithm is an estimate of the frequency spectrum corresponding to each particular kind of drum. The W matrix will be referred to as the templates matrix. The H matrix corresponds to the activations matrix, which shows the period of time when the templates appear in the mix. The H matrix thus identifies when a specific drum is being played in the recording. The V matrix corresponds to the STFT of the input recording. Each time NMF is run in this investigation, the matrices are updated until two consecutive iterations produce matrix estimates that differ in no entry by 0.0005 in magnitude or more. In this investigation, when NMF is run, only one of the matrices – either H or W – is updated at a time. NMF was used, in part, because in many research investigations it has been successful in accomplishing the similar task of automatic drum transcription [2].

B. Block Diagram

A Block Diagram for the overall system is shown in Fig. 2. NMF is run in two distinct cases to accomplish track separation, each time only updating one of the matrices, W or H . Initially, in this investigation, both matrices were updated at once. However, separating the updating of each matrix allowed

for the use of different initialized guesses for the W and H matrices which produced more accurate output matrices.

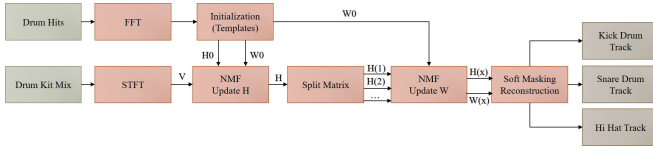


Fig. 2. System Block Diagram

1) *Templates Initialization:* The templates matrix is approximated by averaging the frequency spectra of 20 different drum hits for each kind of drum being identified (kick, snare, and hi-hat). These templates are smoothed using a 3-point moving average filter. For each template, if the magnitude in a frequency bin is below an order-of-magnitude of the maximum template magnitude, the magnitude of said frequency bin is set to zero (Fig. 3). This thresholding allows for a faster NMF performance, with less risk of error. This thresholding method was able to achieve higher true positive and lower false positive scores (testing methods are explained in the Results section). In the second iteration of NMF, when the templates matrix is being updated, the non-thresholded matrix (Fig. 4) is used, which allows the algorithm to more accurately approximate the drum templates.

2) *NMF Run 1: Activation's Updating:* The first time NMF is performed, it is used to generate an accurate activations matrix. The thresholded templates matrix is provided, which estimates the frequency spectrum of each particular kind of drum. The algorithm is run with a randomized (non-zero and non-negative) initial activations matrix. The magnitude STFT spectrogram of the input recording is also supplied. The output activations matrix (Fig. 5) of this first NMF sequence predicts where the templates occur in the song, or when the specific drums are being played.

3) *Array splitting:* To generate improved reconstruction, the activations and templates matrices were altered to account for specifics of drum sounds. First, the templates and activations matrices were both expanded to account for the sound of the onset of each drum. For the templates matrix, the template of each kind of drum was merely copied into an additional column of the matrix (Fig. 6). For the activations matrix, when a drum hit was detected, the first four frames (20 ms) were copied into an additional row of the matrix and deleted from the original row (Fig. 7).

It is presumed that each time a drum hit occurs, the resulting frequency spectrum of the sound produced will be slightly different. For example, if the drummer hits the center of a drum (or cymbal) the sound will be significantly different than when she hits closer to the rim. To account for these changes, the activations matrix is split into blocks and NMF is run, updating the templates matrix, on each block (Fig. 8). This allows the template of each drum hit to update individually. The activations matrix was split

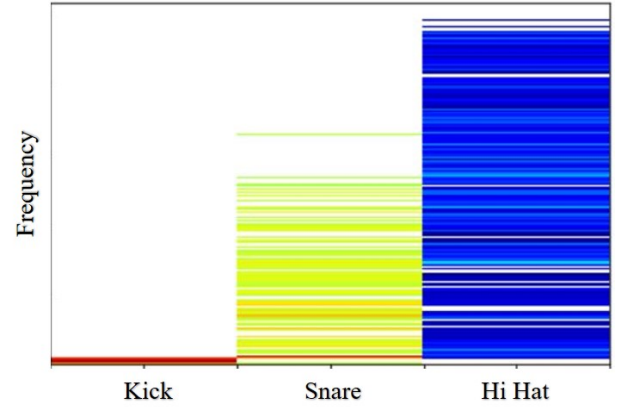


Fig. 3. Templates Matrix: Thresholded

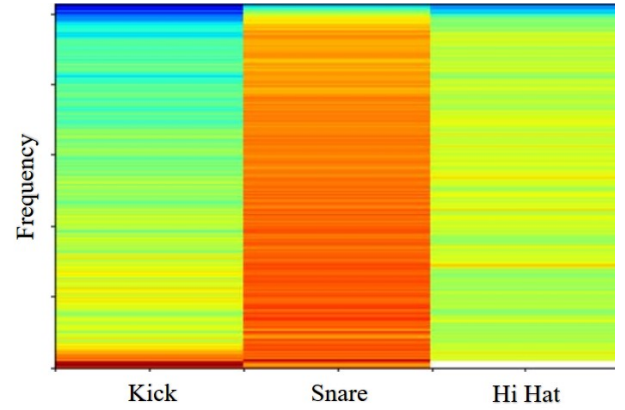


Fig. 4. Templates Matrix: Un-Thresholded

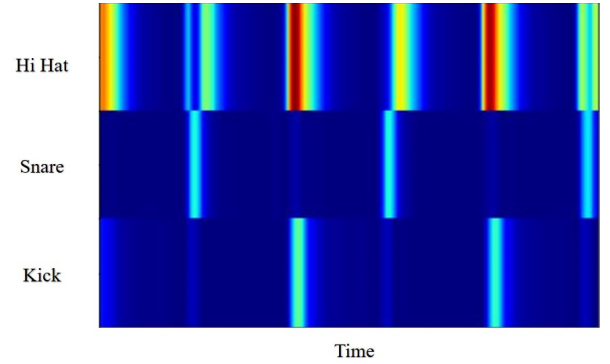


Fig. 5. Activations Matrix

into sections corresponding to when a drum hit was no longer detected. With this method, when two drums were detected at the same time both hits were included in one block.

4) *NMF Run 2: Template Updating:* The second time NMF is run, it is used to generate a more accurate templates matrix. In this NMF run, only the templates matrix is updated. For each computed drum hit (or set of different drum hits)

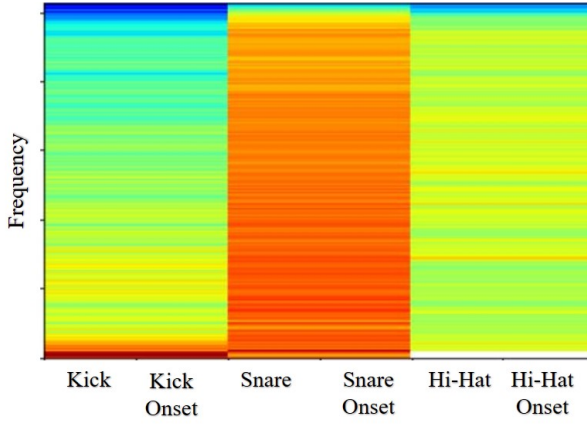


Fig. 6. Templates Matrix with Onsets

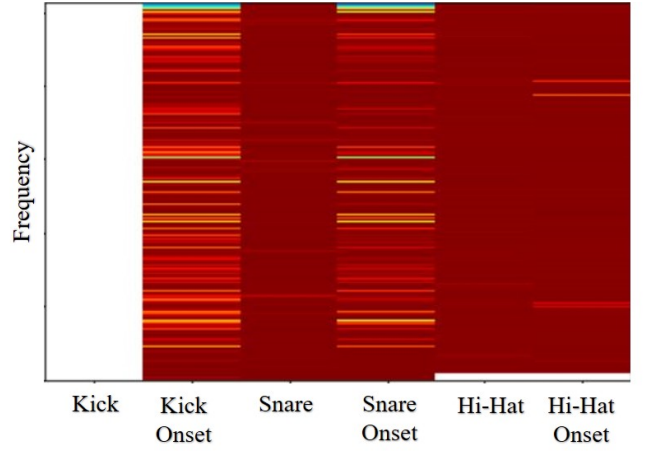


Fig. 9. Output Templates Matrix

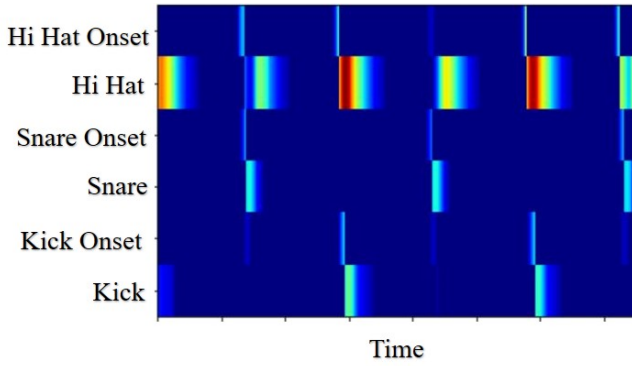


Fig. 7. Activations Matrix with Onsets

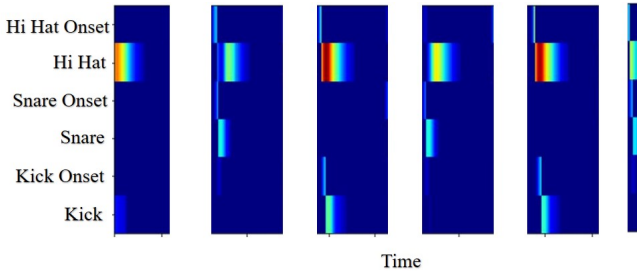


Fig. 8. Separated Activations Matrix

separated as described above, NMF is used to generate the drum spectrum for that hit. This run of NMF is supplied with the initialized templates, without thresholding, to allow for a more flexible output (Fig. 4). The lowest 20 frequency bins of the hi-hat template were initialized to zero. This was done given that most of the hi hat spectral energy is above these bottom bins and that this zeroing-out reduces the overlap between the hi hat and kick drum spectra, yielding more isolated results. Onset templates are also provided using copies of the corresponding drum's template. The NMF algorithm is also supplied with the input activations matrix and the input audio mix STFT. This activations matrix is

thresholded such that any matrix entry below a threshold of 7 is set to zero. The output of this NMF run is a templates matrix with a spectrum for each kind of drum hit struck during a specific block of time.

5) *Soft Masking Reconstruction*: The last step in the block diagram is to utilize the generated templates and activations matrices to reconstruct the separated tracks of audio. For each separated block, the activations that do not correspond to a specific drum are zeroed out. This forms a drum-specific activations matrix H^x (where x denotes one of the three drum types). This isolated version is multiplied by the templates matrix and is then used to construct a drum specific mask M^x . This mask is multiplied by the corresponding time section of the original STFT, diminishing the components that do not correspond to the right drum being played.

$$M^x = H^x W \oslash (WH + \epsilon)$$

These masks are then applied to the STFT, χ to form the approximated drum-specific spectrogram $\hat{\chi}^x$.

$$\hat{\chi}^x = M^x \otimes \chi$$

Finally, the ISTFT is taken of $\hat{\chi}^x$, which generates an audio track for the given drum type for a specific segment of time. These tracks are then segmented together to create the isolated drum audio tracks.

III. RESULTS

The proposed method was assessed using the output activations matrix. When an element of the matrix was above a magnitude of 7 it was interpreted as a positive identification of a particular drum being played. Consecutive positive identifications were grouped together into blocks. Two statistics were used to gauge the identification performance. The first was a true positive statistic corresponding to the percent of drum onsets that were correctly identified at

the correct time. The second statistic was a false positive statistic calculated by summing the number of groups of hits identified in the activations matrix that corresponded to real drum hit. The algorithm was run on the IDMT-SMT Drums dataset, which provides annotated times that a drum is hit and the corresponding type of drum being played at that given time [3]. This dataset contains 93 mixed drum audio clips with a hi-hat, snare drum, and kick drum. The clips have a mix of acoustically recorded audio, wave drum audio, and synthesized audio. These annotations were used to generate the true positive and false positive statistics, indicating when the activations matrix could correctly identify drums onsets and when it incorrectly identified onsets that did not correspond to an annotation. When run on the entire IDMT-SMT Drums mixed drum dataset the algorithm achieved a true positive identification score of 96 percent and a false positive identification score of 21 percent.

IV. DISCUSSION

While the achieved identification scores are promising, some additional investigation could achieve better results. There seems to be a trade-off with regard to the threshold magnitude chosen to identify a positive hit, with a lower threshold yielding a better true positive score and a higher threshold yielding a better false positive score. More nuanced approaches to identification may be able to better account for both measurement statistics. These identification scores also fail to gauge the performance of the reconstruction component of the algorithm. Another metric may be able to be applied to gauge the similarity to an input recording of an isolated drum track mixed into a recording with multiple drums.

One of the audible issues with the reconstruction is that when two drums are played at once, there is often some spectral bleed between them so you might get some higher frequency components in a kick drum track when it is played at the same time as the hi-hat. Additional experimentation could be done to identify how to better isolate drums played at the same time.

An extension of this project could also be done to extend to include other drum types. A potential failing of the algorithm would likely occur if a drum is searched for that is not present in the mix or if there is a drum in the input audio mix that does not have its own template. Other issues may arise when this problem is expanded given the more overlapping spectra of other drum types, which may yield degraded performance.

REFERENCES

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- [2] Wu, Chih-Wei & Dittmar, Christian & Southall, Carl & Vogl, Richard & Widmer, Gerhard & Hockman, Jason & Müller, Meinard & Lerch, Alexander. (2018). A review of automatic drum transcription. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*. PP. 1-1. 10.1109/TASLP.2018.2830113.

- [3] Christian Dittmar, Daniel Gärtner: Real-time transcription and separation of drum recordings based on NMF decomposition, *Proceedings of the 17th International Conference on Digital Audio Effects (DAFx-14)*, Erlangen, Germany, 2014